SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I K. Bhavya Sri/192472245 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Bhavya

Descriptive Written Test

1. Intelligent Agents in a Smart Home Automation System

Introduction

An intelligent agent observes its environment through sensors and takes actions to achieve desired results. In a smart home, it controls lighting, temperature, and security based on user preferences and environmental conditions.

Types of Intelligent Agents

1. Simple Reflex Agent

- Acts based on the current input using predefined rules.
- Example: Turns on the light when motion is detected.
- Advantage: Fast and simple.
- Disadvantage: Cannot remember past events or learn user preferences.

2. Model-Based Agent

- Maintains an internal model of the environment using current and previous information.
- Example: Remembers that someone is inside the house even when the motion sensor temporarily detects no movement.
- Advantage: Makes better decisions using stored information.
- Disadvantage: Requires more memory and processing.

3. Goal-Based Agent

- Selects actions to achieve specific goals.
- Example: Adjusts the temperature to reach the user's preferred room temperature.
- Advantage: Plans actions to achieve desired outcomes.
- Disadvantage: May require more time for planning.

4. Utility-Based Agent

- Selects actions based on a utility function that evaluates comfort, energy consumption, and security.
- Example: Adjusts air conditioning to maintain comfort while reducing electricity usage.
- Advantage: Balances multiple objectives effectively.
- Disadvantage: Designing a suitable utility function can be complex.

Most Effective Approach

A hybrid model-based and utility-based agent, combined with goal-based planning, would be most effective.

The model-based component remembers environmental conditions and user preferences. The goal-based component plans actions, while the utility-based component balances comfort, energy efficiency, and security. Learning mechanisms can update user preferences over time.

Conclusion

A hybrid intelligent agent provides better adaptability, comfort, energy savings, and security than relying on a simple reflex agent alone.

2. Solving an Unknown Maze Using UCS and IDS

Introduction

A robot in an unknown maze must explore different paths to reach the exit. Uninformed search algorithms explore the maze without using additional knowledge about the direction of the goal.

1. Uniform Cost Search (UCS)

- UCS expands the path with the lowest total cost first.
- The robot records the cost of each path and explores the least expensive option.
- It finds the optimal solution when path costs are non-negative and the search conditions are appropriate.

Advantages:

- Finds the lowest-cost path.
- Handles paths with different movement costs.

Disadvantages:

- Can consume considerable memory.
- May take a long time when many low-cost paths exist.

Complexity: Time and space complexity are generally exponential in the worst case, depending on path costs and the minimum step cost.

2. Iterative Deepening Search (IDS)

- IDS performs depth-limited searches, starting at depth 0 and increasing the limit gradually.

- The robot explores paths up to the current depth and repeats the search with a larger limit until it finds the exit.

Advantages:

- Uses less memory than UCS.
- Finds the shallowest solution when all steps have equal cost.
- Suitable for large search spaces.

Disadvantages:

- Repeats some exploration at each depth limit.
- Does not guarantee the lowest-cost path when movement costs differ.

Complexity: Time is \(O(b^d)\), and space is \(O(bd)\), where \(b\) is the branching factor and \(d\) is the depth of the shallowest solution.

Comparison

Feature	UCS	IDS
Search method	Lowest-cost path first	Increasing depth limits
Optimality	Yes, for non-negative costs under standard conditions	Yes, for equal step costs
Time	Can be high	\(O(b^d)\)
Space	High	\(O(bd)\)
Best suited for	Different movement costs	Limited memory and equal movement costs

Relation to Intelligent Agents

- Simple reflex agent: Chooses actions using immediate sensor inputs but may get trapped in dead ends.
- Model-based agent: Remembers visited locations and obstacles to improve decisions.
- Goal-based agent: Plans a path toward the maze exit.
- Utility-based agent: Chooses a path by balancing distance, time, and risk.

Most Effective Approach

A model-based, goal-based agent using IDS is suitable when the maze has equal movement costs and the robot has limited memory. The model-based agent records

discovered walls and visited locations, while the goal-based agent searches for the exit.

If different paths have different costs, UCS with a model-based, goal-directed agent is preferable because it can find the least-cost route.

Conclusion

The best combination depends on the maze. IDS is memory-efficient for equal-cost paths, whereas UCS is preferable when finding the cheapest path is important. Combining search with a model-based agent improves the robot's ability to explore and make informed decisions.